

JUNKAI JI

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EDUCATION

Politecnico di Milano

Master of Science in **High Performance Computing Engineering**,

Milan, Italy
Sep 2025 – Present

Università di Bologna

Bachelor of Science in **Computer Science and Engineering**,

Cesena, Italy
Sep 2021 – Nov 2024

EXPERIENCE

Supply Chain Backend Engineer [Internship]

Accenture

- **Incoming**

Sep 2026 – Mar 2027

Milan, Italy

Web Backend Developer [Full-time]

Cosmobile

- Designed and delivered RESTful APIs using **NestJS** and **TypeScript**
- Developed a Python microservice for automated PDF catalog generation
- Built a RAG pipeline over OpenAI for voice-driven product search and recognition
- Created **OrderSender**, a Zapier plugin that automates order purchasing
- Deployed **Uptime Kuma** on **GCP** for service uptime monitoring

Aug 2024 – Aug 2025

Bologna, Italy

PROJECTS

Patente Plus

TypeScript, Next.js, Cloudflare, Stripe | patenteplus.com

- Full-stack SaaS for Italian driving license prep with Stripe payments and SSO auth
- Uses an LLM API to explain wrong quiz answers; deployed on Cloudflare Workers with SQLite

Hybrid Matrix-Free Solver

C++17, deal.II, MPI | github.com/swampholyten/hybrid-matrix-free-solver

- Hybrid MPI matrix-free finite element solver with geometric multigrid preconditioning

Prequal-rs

Rust, Tokio, Axum | github.com/swampholyten/prequal-rs

- Reproduction of Google's PreQual load-balancing policy (NSDI 2024)

AstraLog-HPC

C++23, OpenMP

- Rule-based anomaly detector for spacecraft telemetry
- Implemented a CI/CD pipeline with GitHub Actions

Lox Interpreter

Rust

- Tree-walk interpreter for the Lox language: scanner, recursive-descent parser, and evaluator

TECHNICAL SKILLS

Languages: Rust, C++, C, TypeScript, Python

Frameworks: Next.js, Node.js, NestJS, Axum

Parallel & GPU: CUDA, OpenMP, MPI

Databases: PostgreSQL, SQLite, Redis

Expertise: Web & Parallel Programming, Data Structures & Algorithms, OOP

Dev Tools: Docker, Git, Linux

Spoken Languages: Chinese (Native), Italian (Fluent), English (Fluent)